

Feedback Agent Walkthrough

Overview

The Feedback Agent has been successfully developed to handle user corrections securely and preserve them effectively for future retraining. By maintaining its own image repository, it remains robust even if users delete their original uploads.

Changes Made

- **Feedback Agent Implementation:**
 - Located at [feedback_agent.py] (file:///d:/agricare/agents/feedback_agent/feedback_agent.py).
 - Maintains `data/feedback/images/` to copy and secure all feedback images immediately upon submission.
 - Utilizes a robust CSV schema (`data/feedback/feedback_log.csv`) incorporating:
`image_path,predicted_class,actual_class,confidence,timestamp,model_version`
- **Feature Additions:**
 - **Duplicate Detection:** Computes an MD5 hash of incoming images. If the hash already exists as a filename in `data/feedback/images/` , it blocks the entry to prevent noisy duplicates.
 - **Statistics Generation (`get_feedback_stats`):** Instantly returns `total_feedback` , `most_confused_class` (most frequent correct class that the model got wrong), and `average_wrong_confidence` .
 - **Retraining Exporter (`export_retraining_dataset`):** Iterates through the feedback logs and systematically organizes images into `data/retraining_dataset/{actual_class}/image.jpg` so they are immediately usable by data generators.
- **Unit Testing:**
 - Comprehensive lifecycle test added in [test_feedback_agent.py] (file:///d:/agricare/tests/test_feedback_agent.py).

Validation Results

- Ran `venv\Scripts\python -m unittest tests/test_feedback_agent.py` .
- Ran 1 test in 0.063s — Execution verified the complete pipeline successfully:
 - Initial log copies the image and registers the CSV entry.
 - Duplicate submissions are successfully blocked.
 - Stats reflect the single dummy entry correctly.
 - The retraining exporter creates the class sub-folder and copies the image securely.